

Exponential Form and Logarithmic Form

Answer Key

Algebra 2

Quick Answers

- | | |
|---|---|
| 1. 5 | 6. -5 |
| 2. 125 | 7. $\frac{1}{2}$ |
| 3. 3 | 8. 12 hours |
| 4. (a) the value of the power = $\frac{1}{49}$, (b) the logarithm = -2 | 9. 6 |
| 5. $\frac{1}{9}$ | 10. (a) the exact solution = $\frac{17}{3}$, (b) the solution as a decimal = 5.67, — |
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What is verified

12 of 13 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSF-BF.B.5 – *problems 6, 10*

HSF-LE.A.4 – *problems 1, 2, 3, 4, 5, 7, 8, 9*

Difficulty 1–5 of 5 across 13 tagged checks.

- 1.** Rewrite $2^5 = 32$ in logarithmic form and give $\log_2 32$.

Solution: The base 2 stays the base; the exponent 5 becomes the value of the logarithm and 32 becomes its input: $2^5 = 32 \iff \log_2 32 = 5$.

$\log_2 32 = 5$

- 2.** $\log_5 x = 3$: rewrite in exponential form and give x .

Solution: A logarithm is an exponent, so $\log_5 x = 3$ says “5 to the power 3 gives x ”: $5^3 = x$, and $5^3 = 125$.

$x = 125$

- 3.** Evaluate $\log_4 64$.

Solution: The question is “which power of 4 gives 64?” $4^1 = 4$, $4^2 = 16$, $4^3 = 64$.

$\log_4 64 = 3$

4. (a) 7^{-2} as a fraction. (b) $\log_7(1/49)$.

Solution (a): A negative exponent means the reciprocal of the positive power: $7^{-2} = \frac{1}{7^2} = \frac{1}{49}$. $\frac{1}{49}$

Solution (b): Part (a) is the exponential form of exactly the statement being asked about, so switching it round gives $\log_7\left(\frac{1}{49}\right) = -2$. A logarithm may be negative — what may not be negative is its *input*. -2

5. $\log_3 x = -2$: exponential form and x .

Solution: $\log_3 x = -2$ becomes $3^{-2} = x$, and $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$.

$x = \frac{1}{9}$

6. Solve $2^x = \frac{1}{32}$.

Solution: Write the right side as a power of the same base: $\frac{1}{32} = \frac{1}{2^5} = 2^{-5}$. Equal bases force equal exponents. (In logarithmic form the same statement reads $\log_2 \frac{1}{32} = x$.)

$x = -5$

7. Evaluate $\log_9 3$.

Solution: Which power of 9 gives 3? Since $\sqrt{9} = 3$ and a square root is the one-half power, $9^{1/2} = 3$.

$\log_9 3 = \frac{1}{2}$

8. Doubling culture $N = 2^t$, $N = 4096$. Find t .

Solution: The count statement $2^t = 4096$ becomes $t = \log_2 4096$ in logarithmic form. Counting doublings: $2^{10} = 1024$, $2^{11} = 2048$, $2^{12} = 4096$.

$t = 12$ hours

9. $\log_b 216 = 3$: find the base b .

Solution: Switching form gives $b^3 = 216$. Here the unknown is the *base*, so undo the cube rather than a logarithm: $b = \sqrt[3]{216}$, and $6^3 = 216$. A base must be positive, so $b = 6$.

$b = 6$

10. Challenge: $\log_2(3x - 1) = 4$. (a) Exact x . (b) Decimal. (c) Why $x \leq \frac{1}{3}$ is impossible.

Solution (a): Switch to exponential form, then solve the linear equation:

$$2^4 = 3x - 1$$

$$16 = 3x - 1$$

$$3x = 17.$$

$$x = \frac{17}{3}$$

$$5.67$$

Solution (b): $\frac{17}{3} = 5.666\dots$

Solution (c): *Instructor-judged.* The exponential form says $3x - 1 = 2^4$, and every power of 2 is positive, so $3x - 1$ must be positive — that is, $x > \frac{1}{3}$. Equivalently, a logarithm is only defined for a positive input, so $\log_2(3x - 1)$ does not exist at all once $3x - 1 \leq 0$. Full credit needs the positivity of the input, tied to either form.